PRP – Analysis & Comparison of LLM Readability Scores

Key Findings

- LLMs perform very well for binary criteria (when something is either black or white, without any in-between).
- For intermediate criteria, various Claude iterations showed that the justifications for each criterion tended to be very similar, yet these led to disparate scores.
- The LLMs made several calculation errors (when converting subscores into a global score or applying the /3 penalty).

#########

Introduction

The aim of this document is to provide an analysis of the quality of the readability scores delivered by Claude based on the provided prompt. To do so, we proceed in 5 steps:

1. We provided Claude with the prompt and 10 text samples to analyze three times.
2. We compared the three iterations generated by Claude to evaluate its consistency and reliability.
3. We provided ChatGPT and Gemini with the same prompt and the same 10 text samples.
4. We analyzed and compared the scores attributed by ChatGPT and Gemini with those produced by Claude. Here again, the purpose is to evaluate the quality of the prompt and Claude's reliability.
5. Based on the previous results, we propose some improvements to the approach and the prompt.

The detailed results delivered by the different LLMs are available HERE.

The LLMs chats are available via the following links:

- Claude1
- Claude2
- Claude3
- Chat GPT
- Gemini

On the next following page, you'll find the prompt and ten text samples used for this exercise.

Prompt used

Return the attached ordered list of projects, ranking those according to their readability, on the following scale:

a. Excellent readability: 91% to 100%.
b. Good readability: 75% to 90%.
c. Just readable: 66% to 75%.
d. Difficult to read: 51% to 65%.
e. Cryptic: 0% to 50%.

Use the following weighing criteria:

a. **Desc_EN**: Description is written in English: 25%
b. **Title_EN**: Title is written in English (15%) or is written in another language (0%)
c. **Substantiality**: Title and description altogether give a concrete idea of the project's specific goals, objectives, or activities: 20%
d. **Simplicity**: Both Title and description have a good enough readability for a person with a secondary school degree who do not have English as first language: 10% (note: when the description is not in English, calculate the score then divide it by three, rounding up to the next integer)
e. **Title_clear**: Title should be meaningful and does not contain unexplained acronyms (an acronym is considered explained if it is immediately followed by its full form in parentheses, such as 'Citizen Science (CS)' : 10% (note: when the title is not in English, calculate the score then divide it by three, rounding up to the next integer)
f. **Desc__funding**: Description fields clearly lets understand for which specific purpose funding was allocated: 10% (note: when the description is not in English, calculate the score then divide it by three, rounding up to the next integer)
g. **Title_standalone**: Title makes sense without reading the description: 10% (note: when the title is not in English, calculate the score then divide it by three, rounding up to the next integer)

For each project, provide a score card in the following format, replacing the content in brackets:

- Project: [record number + title_text]
- Desc: [desc_tewt]
- ID: [ID_num]
- [ranking on scale]: [total score]%
    - Desc_EN: [subscore for criterion] out of 25%: [justification]
    - Title_EN: [subscore for criterion] out of 15%: [justification]

    - Substantiality: [subscore for criterion] out of 20%: [justification]
    - Substantiality: [subscore for criterion] out of 20%: [justification]
    - Simplicity: [subscore for criterion] out of 10%: [justification]
    - Title_clear: [subscore for criterion] out of 10%: [justification]
    - Title_standalone: [subscore for criterion] out of 10%: [justification]

Proceed one project at a time and ask for my feedback at each iteration.

Notes

- The analysis only considers the first 200 characters of the description.
- After observing that Gemini tended to give very extreme scores, we added to the prompt that intermediary scores were allowed (except for the first two criteria: Desc_EN and Title_EN).
- The different LLMs made **calculation errors** when providing the overall scores. When this was pointed out, they corrected themselves, but this highlights the need for a control mechanism to verify the calculation of the different subscores into a global score

Ten Text Samples

Text 1

Title: Citizen-enhanced Open Science in Southeastern Europe Higher Education knowledge hubs

Desc: The European Union promotes accountability, transparency, and sustainability in addressing global challenges. Citizen Science (CS) is a vital part of Open Science (OS) and has engaged millions of ...

ID: AAAgrz

Text 2

Title: Stötta och uppmuntra socialt entreprenörskap inom EU

Desc: YBC is a high school in Nacka, just outside of Stockholm, Sweden. The school offers Natural Science, Social Science, and Economic Programs and within the Economics program, we have been running co...

ID: BBBsd

Text 3

Title: När klassrummet flyttar ut i Europas verklighet. - Ett projekt om migration, aktivt medborgarskap och engagerade ungdomar.

Desc: At Fryshuset the voices and perspectives of the youth must be in focus, regardless of who you are or where you come from. To be able to speak up it takes not only knowledge but also self confidenc...

ID: fzgreahh

Text 4

Title: Next Generation Grand Tour through Europe

Desc: As an ÖKOLOG-, MINT-, UNESCO-, BIGling- and voXmi-certified school, it is highly important to us that our students as well as our teaching staff engages with relevant and current topics such as su...

ID: zrhyteiut

Text 5

Title: Accredited projects for mobility of learners and staff in adult education

Desc: Accredited projects for mobility of learners and staff are implemented by organisations holding an Erasmus accreditation. Each accredited project represents a step in the implementation of the org...

ID: zaert

Text 6

Title: Accredited projects for mobility of learners and staff in vocational education and trainings

Desc: Accredited projects for mobility of learners and staff are implemented by organisations holding an Erasmus accreditation. Each accredited project represents a step in the implementation of the org...

ID: rtymlj

Text 7

Title: Monsters in Danger of Extinction

Desc: Monsters in Danger of Extinction project (M.I.D.E.) departs from a previous partnership between two eTwinning schools: IES Unamuno BHI (Bilbao / Spain) and Ekpaidetyria Panou (Nafpaktos / Greece) ...

ID: ezrtrrtymlj

Text 8

Title: Digital Energy Efficiency Designers

Desc: Good DEEDs stems from two needs that are widespread in Europe and on which VET training can take action: 1. Need to overcome the cultural approach to digital as something that does not generate c...

ID: iykg

Text 9

Title: Innovating & Building Understanding to Improve Language Development with Information Technology

Desc: In response to Covid19, the iBUILD IT project aimed to provide sustainable solutions for English language teaching (ELT) and educational inspiration for providing high-quality ELT and instruction ...

ID: aezzr

Text 10

Title: Europa in de klas

Desc: In Flanders there is a major need for students in teacher training to strengthen their knowledge and skills as future teachers; teachers experience new challenges in their profession, e.g. less mo...

ID: asqxwuo

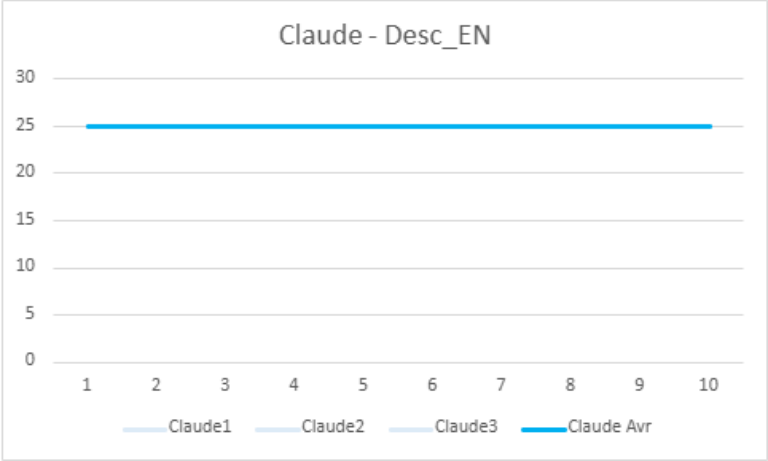
Desc_EN

Introduction criterion

The first weighting criterion introduced by the prompt is ‘Desc_EN’. It refers to whether the description is written in English. It is a yes-or-no criterion: either the full points are awarded, or the score is 0. This criterion represents 25% of the total score.

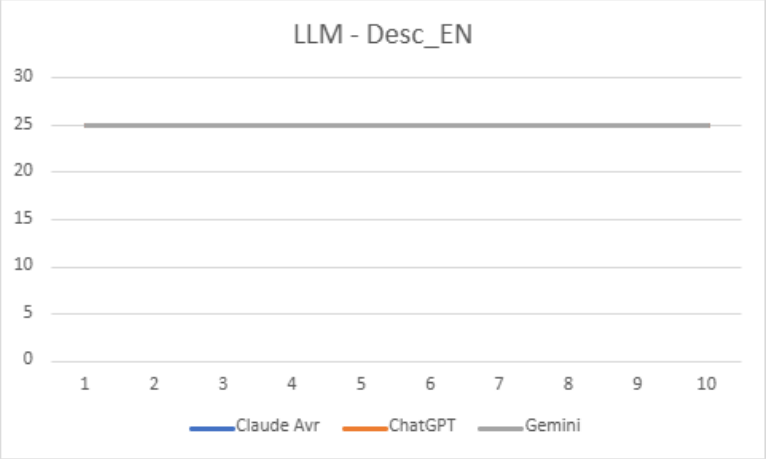
Claude’s results

	Avr	Cl1	Cl2	Cl3	Δ 1 (1-2)	Δ 2 (2-3)	Δ 3 (1-3)
1	**25**	25	25	25	0	0	0
2	**25**	25	25	25	0	0	0
3	**25**	25	25	25	0	0	0
4	**25**	25	25	25	0	0	0
5	**25**	25	25	25	0	0	0
6	**25**	25	25	25	0	0	0
7	**25**	25	25	25	0	0	0
8	**25**	25	25	25	0	0	0
9	**25**	25	25	25	0	0	0
10	**25**	25	25	25	0	0	0



LLM Comparison

	Cl	GPT	Gem	Δ Claude - ChatGPT	Δ Claude - Gemini
1	25	25	25	0	0
2	25	25	25	0	0
3	25	25	25	0	0
4	25	25	25	0	0
5	25	25	25	0	0
6	25	25	25	0	0
7	25	25	25	0	0
8	25	25	25	0	0
9	25	25	25	0	0
10	25	25	25	0	0



Observations

- The three Claude iterations gave the exact same scores for the 10 text samples.
- The three LLMs gave the exact same scores for the 10 text samples.
- Therefore, we can conclude that yes-or-no criteria based on objective scoring (such as whether the text is written in a specific language) can be easily and reliably handled by LLMs.

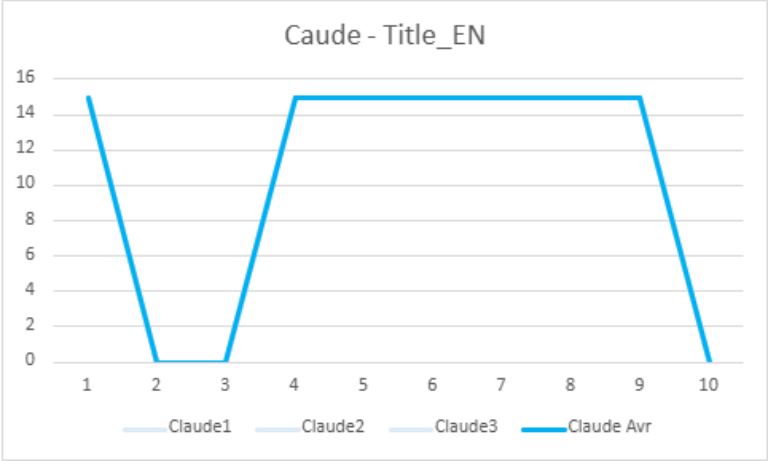
Title_EN

Introduction criterion

The second weighting criterion introduced by the prompt is 'Title_EN'. It refers to whether the title is written in English. It is a yes-or-no criterion: either the full points are awarded, or the score is 0. This criterion represents 15% of the total score.

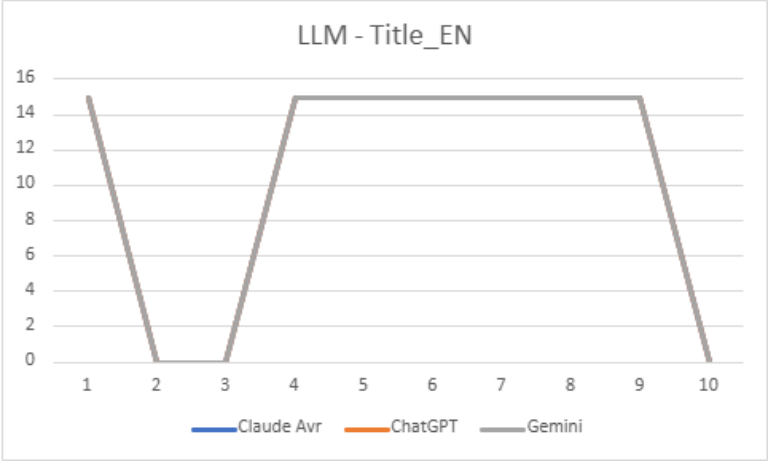
Claude's results

	Avr	Cl1	Cl2	Cl3	Δ 1 (1-2)	Δ 2 (2-3)	Δ 3 (1-3)
1	**15**	15	15	15	0	0	0
2	**0**	0	0	0	0	0	0
3	**0**	0	0	0	0	0	0
4	**15**	15	15	15	0	0	0
5	**15**	15	15	15	0	0	0
6	**15**	15	15	15	0	0	0
7	**15**	15	15	15	0	0	0
8	**15**	15	15	15	0	0	0
9	**15**	15	15	15	0	0	0
10	**0**	0	0	0	0	0	0



LLM Comparison

	Cl	GPT	Gem	Δ Claude - ChatGPT	Δ Claude - Gemini
1	15	15	15	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	15	15	15	0	0
5	15	15	15	0	0
6	15	15	15	0	0
7	15	15	15	0	0
8	15	15	15	0	0
9	15	15	15	0	0
10	0	0	0	0	0



Observations

- The three Claude iterations gave the exact same scores for the 10 text samples.
- The three LLMs gave the exact same scores for the 10 text samples.
- Therefore, we can conclude that yes-or-no criteria based on objective scoring (such as whether the text is written in a specific language) can be easily and reliably handled by LLMs.

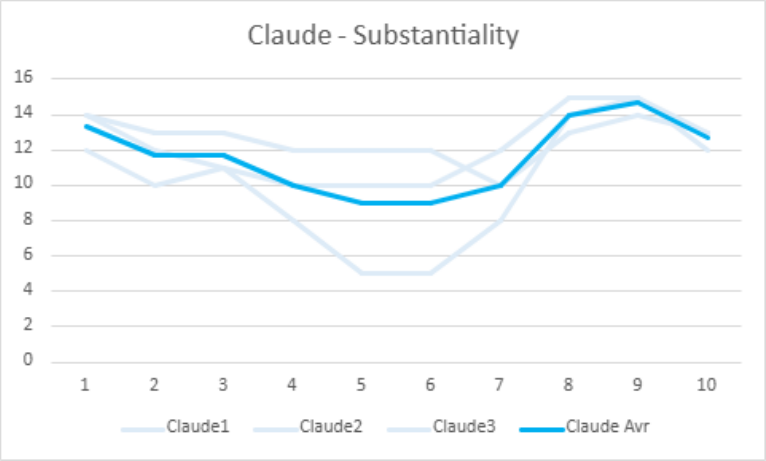
Substantiality

Introduction criterion

The third weighting criterion introduced by the prompt is 'Substantiality'. It refers to whether the title and description together provide a concrete idea of the project's specific goals, objectives, or activities. Unlike the previous criteria, this is not a yes-or-no criterion: the LLM may assign any intermediary score between the minimum and the maximum. This criterion represents 20% of the total score.

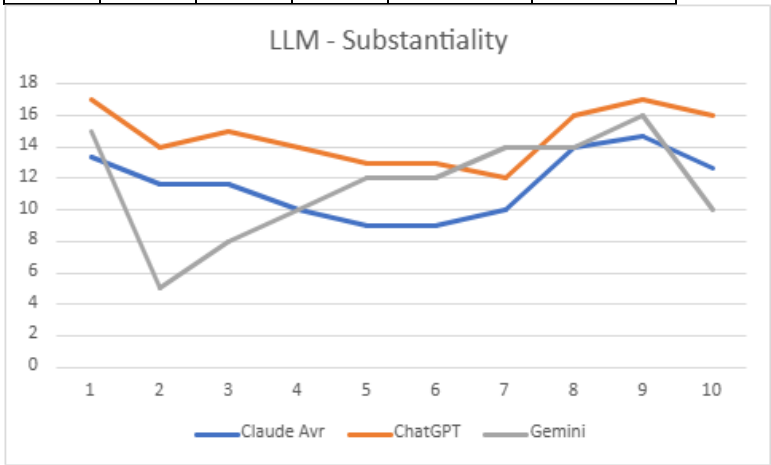
Claude's results

	Avr	Cl1	Cl2	Cl3	Δ 1 (1-2)	Δ 2 (2-3)	Δ 3 (1-3)
1	**13**	12	14	14	2	0	2
2	**12**	10	12	13	2	1	3
3	**12**	11	11	13	0	2	2
4	**10**	8	10	12	2	2	4
5	**9**	5	10	12	5	2	7
6	**9**	5	10	12	5	2	7
7	**10**	8	12	10	4	2	2
8	**14**	14	15	13	1	2	1
9	**15**	15	15	14	0	1	1
10	**13**	12	13	13	1	0	1



LLM Comparison

	Cl	GPT	Gem	Δ Claude - ChatGPT	Δ Claude - Gemini
1	13	17	15	4	2
2	12	14	5	2	7
3	12	15	8	3	4
4	10	14	10	4	0
5	9	13	12	4	3
6	9	13	12	4	3
7	10	12	14	2	4
8	14	16	14	2	0
9	15	17	16	2	1
10	13	16	10	3	3



Observations

We observe significant deltas between the different Claude iterations. Deltas higher than 20% of the category points are indicated in red in the table above. If we look more closely at these high deltas, we can observe that Claude provides consistent commentary on the criterion. Nevertheless, the scores associated with these commentaries can vary significantly.

Let's use text sample 7 as an illustration:

- **Claude 1** attributed a score of **8/20**, justified as follows: *The title is intriguing but ambiguous — it is unclear whether "monsters" is metaphorical or literal. The description establishes a cross-national school partnership (Spain/Greece) via eTwinning, but the truncation prevents understanding of specific goals or activities.*
- **Claude 2** attributed a score of **12/20**, justified as follows: *The title is intriguing but metaphorical — it does not immediately convey the project's actual subject matter. The description clarifies it is a cross-school eTwinning partnership between Spain*

and Greece, but truncation prevents understanding of specific goals or activities. Together they give a partial but incomplete picture.

- **Claude 3** attributed a score of **10/20**, justified as follows: *The title is intriguing but ambiguous — "Monsters" is unclear without context (are these mythological creatures? metaphors?). The description clarifies it is a school partnership project between Spain and Greece, but the truncation prevents understanding of specific goals or activities. The acronym M.I.D.E. is explained in the description, which helps.*

All the 3 commentaries highlight the following elements:

1. Title is intriguing but ambiguous/metaphorical
2. The description establishes a cross-school partnership (Spain/Greece)
3. The truncation prevents good understanding.

Yet, the delta remains high, up to 20% of the category points. The same can be observed for text samples 4, 5 and 6.

When comparing the results from Claude (based on the average of the three iterations) with the results given by ChatGPT and Gemini, we also observe significant deltas. With ChatGPT, this delta can reach up to 20%, while it can reach 33% with Gemini.

The cases with a high delta between Claude's results and ChatGPT's results are similar to the high deltas observed between Claude's own iterations. This means that even if the comments are very similar, they can lead to scores that vary significantly.

Gemini tends to include fewer elements in its analysis than the other LLMs, which might lead to higher deltas. For example, for text samples 2 and 6, it did not include an analysis of the titles alongside the descriptions, whereas the titles were taken into account by Claude and ChatGPT and had an impact on the substantiality score.

These elements highlight the fact that subjective criteria are more difficult for AI to handle. Here, it is not a 'yes-or-no' question but rather a subtle assessment of the overall level of understanding: does the reader gain a concrete idea of the project's specific goals, objectives, or activities?

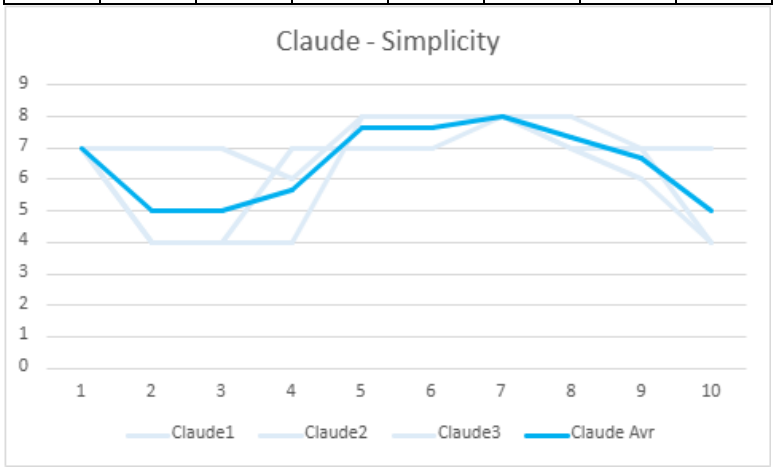
Simplicity

Introduction criterion

The fourth weighting criterion introduced by the prompt is 'Simplicity'. It refers to whether both title and description have a good enough readability for a person with a secondary school degree who do not have English as first language. For this criterion, the LLM may assign any intermediary score between the minimum and the maximum. When the **description** is not in English, it should calculate the score and then divide it by three, (rounded up to the next integer). This criterion represents 10% of the total score.

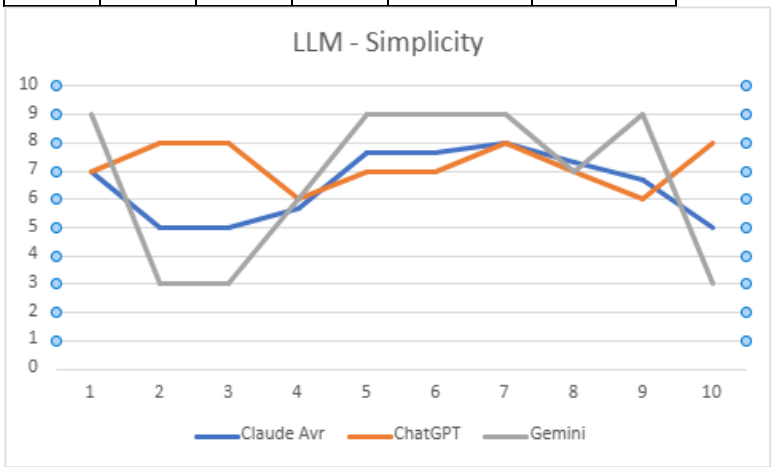
Claude's results

	Avr	Cl1	Cl2	Cl3	Δ 1 (1-2)	Δ 2 (2-3)	Δ 3 (1-3)
1	**7**	7	7	7	0	0	0
2	**5**	4	7	4	3	3	0
3	**5**	4	7	4	3	3	0
4	**6**	4	6	7	2	1	3
5	**8**	8	8	7	0	1	1
6	**8**	8	8	7	0	1	1
7	**8**	8	8	8	0	0	0
8	**7**	7	8	7	1	1	0
9	**7**	7	7	6	0	1	1
10	**5**	4	7	4	3	3	0



LLM Comparison

	Cl	GPT	Gem	Δ Claude - ChatGPT	Δ Claude - Gemini
1	7	7	9	0	2
2	5	8	3	3	2
3	5	8	3	3	2
4	6	6	6	0	0
5	8	7	9	1	1
6	8	7	9	1	1
7	8	8	9	0	1
8	7	7	7	0	0
9	7	6	9	1	2
10	5	8	3	3	2



Observations

Most of the intra-Claude deltas are due to an incorrect application of the prompt. Iterations 1 and 3 of Claude's analysis applied a division of the score by 3 because of a non-English title. However, for the simplicity criterion, this penalty should only be applied in the case of a non-English description.

Examples:

- *Description is in English and reasonably clear. However since the title is not in English, the score is divided by three: full score would be ~7, divided by 3 = 3, rounded up to 4.* (Claude 2, text sample 10)
- *Title is in Swedish and contains no acronyms, and is meaningful in Swedish. Per the rule, base score ~10 → 10/3 = 4 (rounded up).* (Claude 3, text sample 2)

It should also be noted that, for text sample 4, the delta can be explained by the fact that Claude 1 and Claude 2 applied a penalty for the use of unexplained acronyms, whereas Claude 3 did not.

The deltas between Claude and ChatGPT can all be explained by the fact that ChatGPT did – correctly – not apply any penalty for non-English titles. Gemini, however, applied the penalty for non-English titles. Gemini also tends to assign slightly higher scores than Claude.

Possible optimalizations

The prompt should be rewritten to ensure that the penalty is not applied in the case of non-English titles, but only for non-English descriptions.

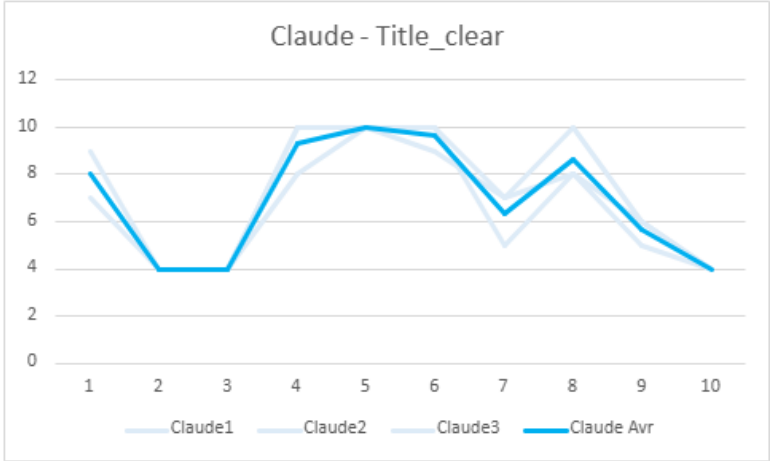
Title_clear

Introduction criterion

The fifth weighting criterion introduced by the prompt is ‘Title_clear’. It refers to whether the title is meaningful and does not contain unexplained acronyms. An acronym is considered explained if it is immediately followed by its full form in parentheses, such as 'Citizen Science (CS)'. For this criterion, the LLM may assign any intermediary score between the minimum and the maximum. When the **title** is not in English, it should calculate the score and then divide it by three, (rounded up to the next integer). This criterion represents 10% of the total score.

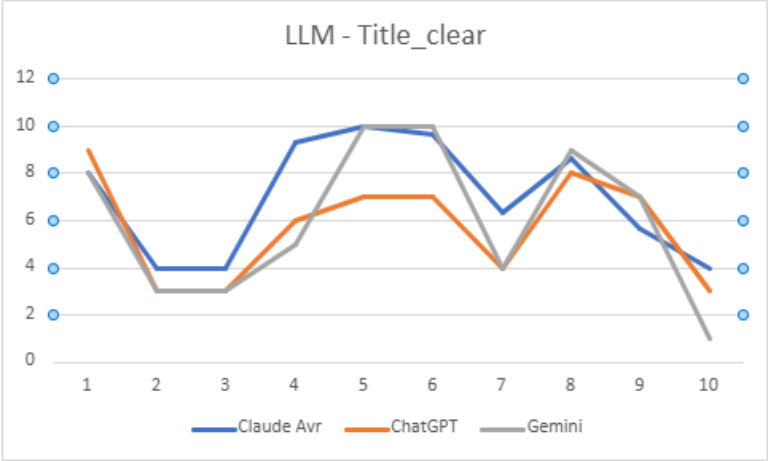
Claude’s results

	Avr	Cl1	Cl2	Cl3	Δ 1 (1-2)	Δ 2 (2-3)	Δ 3 (1-3)
1	**8**	8	9	7	1	2	1
2	**4**	4	4	4	0	0	0
3	**4**	4	4	4	0	0	0
4	**9**	10	8	10	2	2	0
5	**10**	10	10	10	0	0	0
6	**10**	10	9	10	1	1	0
7	**6**	7	7	5	0	2	2
8	**9**	8	10	8	2	2	0
9	**6**	6	6	5	0	1	1
10	**4**	4	4	4	0	0	0



LLM Comparison

	Cl	GPT	Gem	Δ Claude - ChatGPT	Δ Claude - Gemini
1	8	9	8	1	0
2	4	3	3	1	1
3	4	3	3	1	1
4	9	6	5	3	4
5	10	7	10	3	0
6	10	7	10	3	0
7	6	4	4	2	2
8	9	8	9	1	0
9	6	7	7	1	1
10	4	3	1	1	3



Observations

The results of the three Claude iterations are quite similar. Claude's scoring is mostly based on the presence or absence of unexplained acronyms. This is a yes-or-no criterion, which is easily handled by AI.

ChatGPT has taken other factors into account, such as abstractness, cultural references, or the level of genericness, which led to significant deltas in some cases.

Possible optimalizations

We may want to clarify whether this criterion should rely solely on the presence of unexplained acronyms or whether other elements should also be taken into account.

We might also consider applying a minor correction to the prompt, which currently states that *an acronym is considered explained if it is immediately followed by its full form in parentheses, such as 'Citizen Science (CS)'*. In practice, however, it is more common for

the full form to precede the acronym, with the acronym provided afterwards in parentheses.

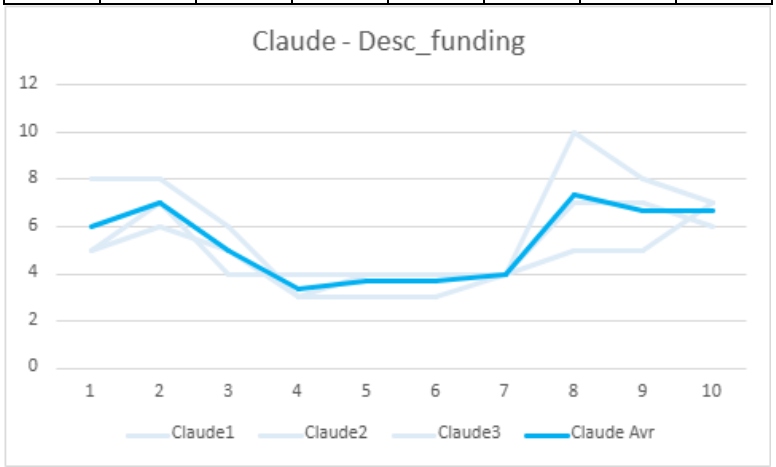
Desc_funding

Introduction criterion

The sixth weighting criterion introduced by the prompt is 'Desc_funding'. It refers to whether the description field clearly indicates the specific purpose for which the funding was allocated. For this criterion, the LLM may assign any intermediary score between the minimum and the maximum. When the **description** is not in English, it should calculate the score and then divide it by three, (rounded up to the next integer). This criterion represents 10% of the total score.

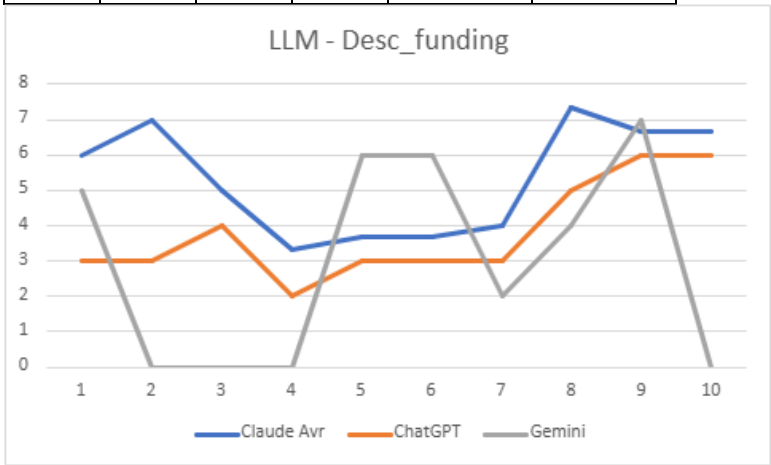
Claude's results

	Avr	Cl1	Cl2	Cl3	Δ 1 (1-2)	Δ 2 (2-3)	Δ 3 (1-3)
1	**6**	8	5	5	3	0	3
2	**7**	8	6	7	2	1	1
3	**5**	6	5	4	1	1	2
4	**3**	3	3	4	0	1	1
5	**4**	3	4	4	1	0	1
6	**4**	3	4	4	1	0	1
7	**4**	4	4	4	0	0	0
8	**7**	10	7	5	3	2	5
9	**7**	8	7	5	1	2	3
10	**7**	7	6	7	1	1	0



LLM Comparison

	Cl	GPT	Gem	Δ Claude - ChatGPT	Δ Claude - Gemini
1	6	3	5	3	1
2	7	3	0	4	7
3	5	4	0	1	5
4	3	2	0	1	3
5	4	3	6	1	2
6	4	3	6	1	2
7	4	3	2	1	2
8	7	5	4	2	3
9	7	6	7	1	0
10	7	6	0	1	7



Observations

Desc_funding is the least stable criterion. Not only do we observe significant deltas among the Claude iterations, but the deltas between the different LLMs can reach up to 70% (by far the highest among all criteria).

This issue likely relies on the fact that some LLMs interpret contextual elements of the project as funding-related information, while others expect explicit references to funding.

At the 2 opposite ends of the spectrum are Claude 1 (which assigns acceptable scores as soon as the project's core activities of goals are mentioned) and Gemini (which assigned multiple scores of 0, stating that the specific purpose of the funding is not mentioned).

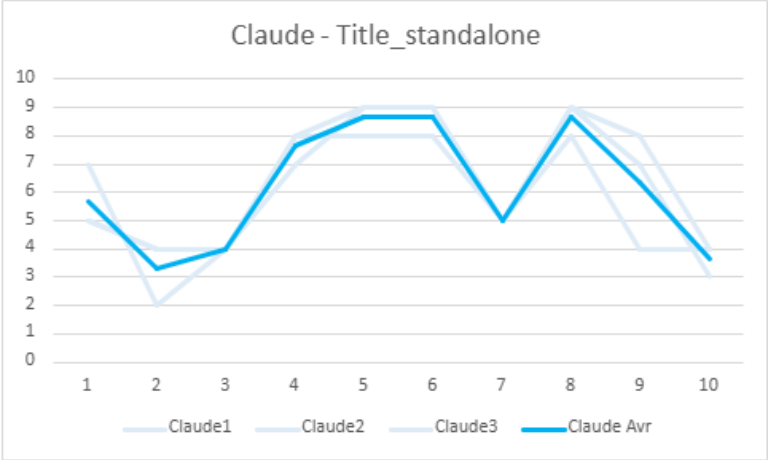
Title_standalone

Introduction criterion

The seveth and last weighting criterion introduced by the prompt is 'Title_standalone'. It refers to whether the title makes sense without reading the description. For this criterion, the LLM may assign any intermediary score between the minimum and the maximum. When the **title** is not in English, it should calculate the score and then divide it by three, (rounded up to the next integer). This criterion represents 10% of the total score.

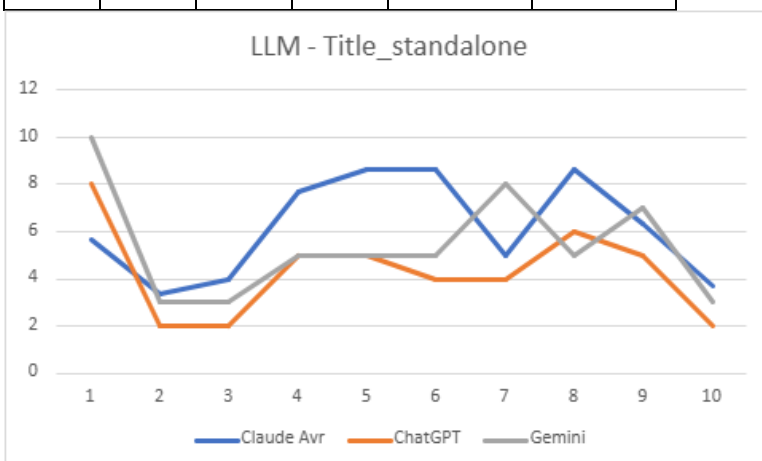
Claude's results

	Avr	Cl1	Cl2	Cl3	Δ 1 (1-2)	Δ 2 (2-3)	Δ 3 (1-3)
1	**6**	7	5	5	2	0	2
2	**3**	2	4	4	2	0	2
3	**4**	4	4	4	0	0	0
4	**8**	8	7	8	1	1	0
5	**9**	9	9	8	0	1	1
6	**9**	9	9	8	0	1	1
7	**5**	5	5	5	0	0	0
8	**9**	9	9	8	0	1	1
9	**6**	7	8	4	1	4	3
10	**4**	3	4	4	1	0	1



LLM Comparison

	Cl	GPT	Gem	Δ Claude - ChatGPT	Δ Claude - Gemini
1	6	8	10	2	4
2	3	2	3	1	0
3	4	2	3	2	1
4	8	5	5	3	3
5	9	5	5	4	4
6	9	4	5	5	4
7	5	4	8	1	3
8	9	6	5	3	4
9	6	5	7	1	1
10	4	2	3	2	1



Observations

Even if the three Claude iterations produce globally similar results, a few challenges remain. These are consistent with what has been observed previously for other criteria.

Similar commentary leading to different scores:

- Text sample 1:
    - **Claude 1** attributed a score of **7/10**, justified as follows: *The title is meaningful on its own, though "knowledge hubs" is vague and slightly jargony without further context.*
    - **Claude 2** attributed a score of **5/10** for **text sample 1**, justified as follows: *The title conveys a geographic and thematic scope but "knowledge hubs" is vague; it doesn't fully stand alone without some context*.
- Text sample 2:

- **Claude 1** attributed a score of **7/10** for **text sample 9**, justified as follows: *The title is meaningful standalone but its length and complexity make it harder to parse quickly for a non-native English speaker*
- **Claude 3** attributed a score of **4/10** for **text sample 9**, justified as follows: *The title is extremely long and while technically self-contained, its complexity undermines standalone clarity. Most readers would struggle to extract the core message without reading the description*.

Calculation errors in the /3 penalty:

- *Title is in Swedish; inaccessible to most. Full score ~7 (fairly descriptive in Swedish), divided by 3 = 3, rounded up to 4*. (Claude 1, text sample 3)

When comparing Claude's results with those from ChatGPT and Gemini, we observe significant deltas. However, these can be explained by the fact that both ChatGPT and Gemini, in addition to evaluating whether the title makes sense, also assessed its quality.

Many of the low scores can be explained by the fact that the LLMs considered that key elements about the project's objectives and activities were missing. This represents a deviation from the prompt, which allows us to conclude that Claude's results were more accurate for criterion 7.

Possible optimalizations

We might consider implementing a control mechanism to verify the calculation of the /3 penalty.

Summary prompt optimization suggestions

General prompt optimizations

- Specify for each criterion whether scores must be binary (maximum or minimum) or can be intermediary (nuanced scores between the minimum and maximum).
- Implement control mechanisms to verify the calculation of subscores into a global score, as well as the correct application of the /3 penalty.
- The more criteria can be broken down into yes-or-no questions, the more reliably AI will be able to handle them.
- Provide the LLMs with human-based analyses to serve as reference points.

Simplicity prompt optimizations

- The prompt should be rewritten to ensure that the penalty is not applied in the case of non-English titles, but only for non-English descriptions.

Title_clear prompt optimizations

- Clarify whether this criterion should rely solely on the presence of unexplained acronyms or whether other elements should also be taken into account.
- Minor correction to the prompt, which currently states that *an acronym is considered explained if it is immediately <u>followed by its full form in parentheses</u>, such as 'Citizen Science (CS)'*. In practice, however, it is more common for the full form to precede the acronym, with the acronym provided afterwards in parentheses.

Next steps

1. Discuss findings with accountable
2. Select 10 new text samples and conduct a human-based self-evaluation
3. Provide the same10 new text samples to Claude
4. Compare the results of step 2 and 3
5. Provide the 10 first text samples along with the human-based evaluation to Claude (3 iterations) & analyze consistency
6. Implement control mechanism for the calculations
7. Adapt the prompt if needed